

Assignment 1 - bonus point

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Exercise 2.1

After completing the baseline network, further steps are taken to improve the performance of the network. For reference, the baseline network attains a test accuracy of 39.24% under the setting $\{\text{lam}=0.1, \text{eta}=0.001, \text{n_batches}=100, \text{n_epochs}=40\}$.

Training-data Expansion

Initially, only a single batch of the dataset was used for training, with another batch reserved for validation. It is natural to expect that incorporating more data would enhance the model's performance. Therefore, while keeping the algorithm and model configuration unchanged, we expand the training set to include all available data, reserving 1,000 samples for validation, with the aim of obtaining a better-trained model.

After conducting this method, the final test accuracy achieved is 41.10%.

Image Flipping

Keeping the size of the training set unchanged, another effective way to strengthen the model is to enrich the diversity of the training data without requiring additional samples, such as horizontally flipping the images.

Under this augmentation strategy, the trained network achieves a test accuracy of 40.24%.

Grid Search

The two strategies above optimize the network at the data level. Alternatively, the network performance can also be improved with better parameters. The grid search is conducted to find a greater combination of amount of L2 regularization(λ), learning rate(η) and batch size(n_batch).

After searching, the optimal setting of parameters are $\{\eta = 0.001, \lambda = 0.1, n_{\text{batch}} = 64\}$, and after 40 epochs of training, the test accuracy is 39.43%.

As shown in Table 1, all methods lead to improvements over the baseline network, although the overall gains remain relatively modest. Among the evaluated methods, training data expansion yields the largest improvement, achieving an increase of approximately 2%.

Table 1: Improvements of network

Strategies	Test Accuracy	Improvement
Raw	39.24%	-
Training-data Expansion	41.10 %	1.86%
Image Flipping	40.24%	1.00%
Grid Search	39.43%	0.19%

Exercise 2.2

Formula Derivation

With the knowledge of

$$l_{\text{multiple bce}}(\mathbf{x}, \mathbf{y}) = -\frac{1}{K} \sum_{k=1}^K [(1 - y_k) \log(1 - p_k) + y_k \log(p_k)]$$

and

$$p = \sigma(s) = \frac{1}{1 + \exp(-s)}$$
$$\frac{\partial p}{\partial s} = \frac{e^{-s}}{(1 + e^{-s})^2} = p \cdot (1 - p)$$

the derivation of $\partial l_{\text{multiple bce}} / \partial s$ is as follows:

$$\begin{aligned} \partial l_{\text{multiple bce}} / \partial s &= \frac{\partial l}{\partial p} \cdot \frac{\partial p}{\partial s} \\ &= -\frac{1}{K} \sum_{k=1}^K \left[\frac{y_k - 1}{1 - p_k} + \frac{y_k}{p_k} \right] \cdot p_k (1 - p_k) \\ &= -\frac{1}{K} \sum_{k=1}^K \frac{y_k - p_k}{p_k (1 - p_k)} \cdot p_k (1 - p_k) \\ &= -\frac{1}{K} \sum_{k=1}^K y_k - p_k \\ &= \frac{1}{K} \sum_{k=1}^K p_k - y_k \end{aligned}$$

Test Accuracy

Our loss function here changes to the multiple binary cross-entropy loss, and the learning rate needs to be multiplied by a factor K as there's a $1/K$ in the loss. The final test accuracy achieved after 40 epochs is then 38.10% with settings $\{\eta = 0.01, \lambda = 0.1, n_{\text{batch}} = 100\}$.

Table 2: Test accuracy of different Loss

Strategy	Parameters	Test Accuracy
Sigmoid + MBCE	$\eta = 0.01, \lambda = 0.1, n_{\text{batch}} = 100$	38.10%
Softmax + CE	$\eta = 0.001, \lambda = 0.1, n_{\text{batch}} = 100$	39.24%

Compared with the cross-entropy method, the MBCE loss method has lower test accuracy.

Ground-truth class Probability

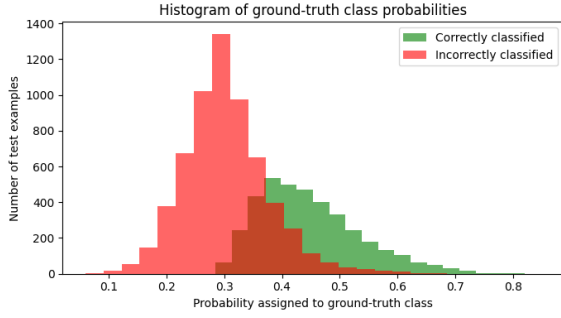


Figure 1: Sigmoid+MBCE

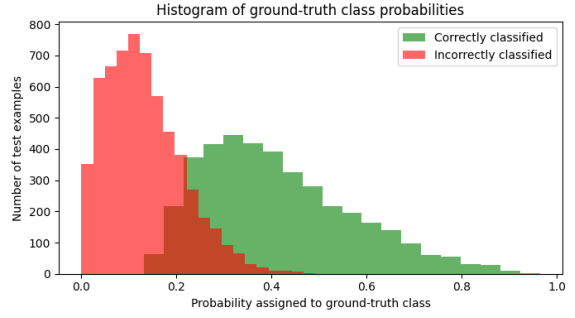


Figure 2: Softmax+CE

As illustrated in the figure, there is a clear qualitative difference between the two histograms. The Softmax + CE method exhibits the following characteristics:

1. Lower decision threshold. For the CE method, the intersection point between correctly and incorrectly classified samples occurs at around 0.15, whereas for MBCE it is closer to 0.3. This indicates that the CE model is able to make correct predictions even when the predicted probabilities are relatively low.
2. Better overall performance From the vertical axis values and the area of the green region, the CE method shows a larger proportion of correctly classified samples. Moreover, the separation between correct and incorrect predictions is less extreme, suggesting a more stable and reliable behavior. In contrast, the MBCE model exhibits higher uncertainty in the probability range of 0.3–0.4, where correct and incorrect predictions are more mixed.

Overall, for this single-label classification task, the Softmax+CE method demonstrates superior performance compared to Sigmoid+MBCE.

Cost and Loss

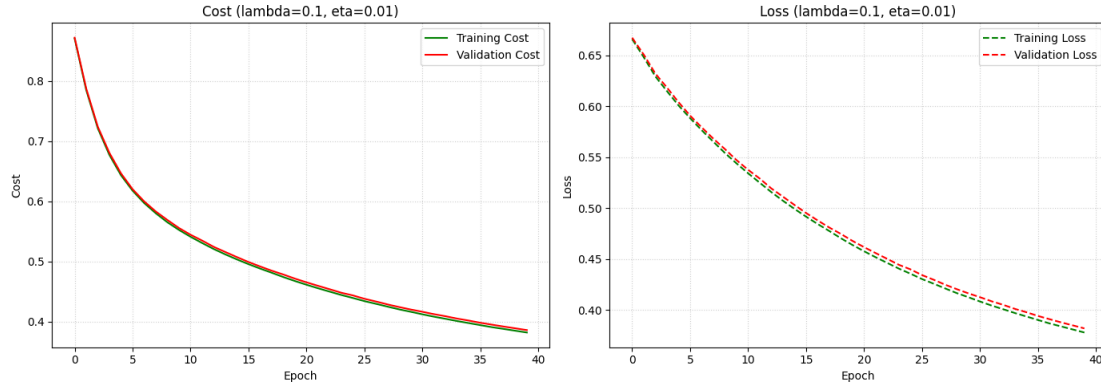


Figure 3: Sigmoid+MBCE

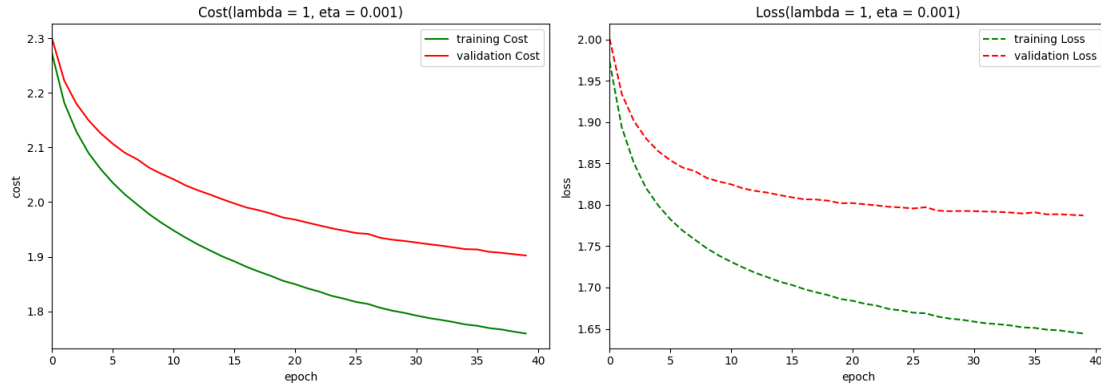


Figure 4: Softmax+CE

Observing the cost and loss graph, the Softmax + CE method exhibits a higher degree of overfitting. In Fig.3, the training and validation curves (red and green) almost completely overlap, indicating strong generalization ability for this method.

In contrast, for the Softmax + CE model (Fig.4), the training loss is consistently lower than the validation loss, and the gap between them increases as the number of epochs grows. This widening gap suggests that the model overfits the training data, leading to poorer generalization performance.